

A self-dual monotone identity is not antisymmetric

Conjecture 00000008845 ziangni-sys

The source classifies self-dual monotone operators by densely defined antisymmetric extensions and antisymmetric bilinear forms. The identity on the real Hilbert line is a counterexample. We explicitly verify self-duality in the convex-Lagrangian sense, rather than relying on the undefined label; we also record its actual adjoint and inverse-graph identities. Antisymmetry here is the skew symmetry of bilinear forms, as in the source's classification, not order-theoretic antisymmetry.

Counterexample. *On $H = \mathbb{R}$, the bounded linear operator $T(x) = x$ is maximal monotone and is generated by a proper continuous convex self-dual Lagrangian. Its graph has no antisymmetric extension. Moreover, T cannot extend any densely defined antisymmetric relation.*

Proof. Consider the everywhere finite quadratic Lagrangian

$$L(x, p) = \frac{x^2 + p^2}{2}.$$

It is continuous, nonnegative, and convex. More explicitly, for $a, b \geq 0$, $a + b = 1$, and $z, w \in \mathbb{R}^2$, its convexity gap is

$$aL(z) + bL(w) - L(az + bw) = \frac{ab}{2}((z_1 - w_1)^2 + (z_2 - w_2)^2) \geq 0.$$

Its full Fenchel conjugate is computed over all $(u, v) \in \mathbb{R}^2$:

$$L^*(q, r) = \sup_{u, v} \{qu + rv - L(u, v)\} = \frac{q^2 + r^2}{2}.$$

Indeed the difference from this upper bound is $-\frac{1}{2}((u - q)^2 + (v - r)^2)$, and equality is attained at (q, r) . Consequently $L^*(p, x) = L(x, p)$: this is genuine self-duality under the exchange of primal and dual variables. The standard equality graph generated by L is

$$\{(x, p) : L(x, p) = xp\} = \{(x, p) : (x - p)^2 = 0\} = \{(x, x) : x \in \mathbb{R}\} = \text{gra } T.$$

This computation includes all dual functionals: every continuous linear functional f on $\mathbb{R}^2$ has the form $f(u, v) = f(1, 0)u + f(0, 1)v$.

Monotonicity is $(x - y)(Tx - Ty) = (x - y)^2 \geq 0$. To check maximality directly, let a monotone relation contain the identity graph and an additional point (x, p) . Test monotonicity against (a, a) with $a = (x + p)/2$. It yields

$$0 \leq (x - a)(p - a) = -\frac{(x - p)^2}{4},$$

so $p = x$. Thus no larger monotone graph exists. The identity is also defined on all of H , satisfies the actual Hilbert adjoint identity $T^* = T$, and its inverse relation equals its graph.

For any antisymmetric relation G , skew symmetry requires

$$\langle p, y \rangle = -\langle x, q \rangle \quad \text{whenever } (x, p), (y, q) \in G.$$

An extension of $\text{gra } T$ contains $(1, 1)$; using that point twice would give $1 = -1$. Conversely, if an antisymmetric relation is contained in $\text{gra } T$, each of its points is (x, x) and the same condition gives $x^2 = -x^2$, hence $x = 0$. Its domain is contained in $\{0\}$ and is therefore not dense in $\mathbb{R}$. This rules out both directions of the terse extension wording. $\square$

The associated form of the identity is $B(x, y) = xy$, which is symmetric and nonzero, not antisymmetric. Thus self-duality and monotonicity cannot imply the asserted skew classification. No failure of the usual theory of self-dual convex Lagrangians is asserted.

Formal verification. The Lean project uses the actual real Hilbert space, continuous linear identity and adjoint. It proves joint convexity and lower semicontinuity of L , and computes the Fenchel supremum for *every* continuous linear functional on the product space, including attainment. It identifies the generated graph, proves its maximal monotonicity and inverse-graph identity, and excludes both antisymmetric extensions and dense-domain antisymmetric restrictions. The pinned Lean 4.19.0 project prints axiom audits of these results.